

# MULTIMEDIA LAB MANUAL

Computer Science - MMV

**Subject:** Introduction of Multimedia

**Duration:** 2 Hours

**Lab Activity:** 01

**Target Audience:** Undergraduate (UG) Students

**Topic:** Structured Data with Pure HTML Tables

## 1. Title & Overview

**Lab Experiment 01:** Editing of a given Structured University Course Timetable & Academic Matrix using Pure HTML Text Elements.

**Objective:** To understand and apply semantic HTML elements for tabular data representation, including complex cell merging (horizontal and vertical alignment), accessible table headings, and organized table sections (<thead>, <tbody>, and <tfoot>) without relying on external CSS layout frameworks or embedded images.

## 2. Problem Statement

As a UG student you are tasked with editing a given well-structured HTML webpage that displays a weekly course timetable for a semester.. You are required to customize this and prepare your time-table.

The solution must meet the following structural requirements:

- Represent 6 working days (Monday to Saturday) across multiple period slots (e.g., 09:00 AM to 04:00 PM).
- Incorporate a consolidated Lunch Break slot that spans across all working days (Horizontal Merging / colspan).
- Incorporate multi-hour lab sessions or continuous blocks that occupy consecutive time slots within a day (Vertical Merging / rowspan).
- Use accessibility attributes (scope="col" and scope="row") on table headers to ensure full screen-reader compliance.
- Provide a summary row in the table footer displaying total academic contact hours per week.
- You are required to change all the colours, fonts, size and font faces style. And Use Sans-Serif Fonts.

## 3. Learning Outcomes

By completing this activity, students will be able to:

1. Construct semantic table hierarchies using <table>, <caption>, <thead>, <tbody>, and <tfoot>.
2. Calculate and execute horizontal cell spanning using the colspan attribute.
3. Calculate and execute vertical cell spanning using the rowspan attribute.
4. Structure clean HTML documents adhering to W3C standards using proper heading tags, text formatting, and tabular data rules.

## 4. Reference

### Step 1: Environment Setup

- Create a working directory named Lab1\_Table\_[YourRollNo].
- Inside this folder, create a new file named timetable.html.

### Step 2: HTML Page Structure

- Set up the basic HTML5 document skeleton (`<!DOCTYPE html>`, `<html>`, `<head>`, `<body>`).
- Add an appropriate `<title>` tag: University Weekly Timetable - [Your Name].
- Add a main heading (`<h1>`) and a brief introductory paragraph describing the academic department and semester.

### Step 3: Table Construction & Semantic Sections

- Create a `<table border="1" cellpadding="8" cellspacing="0">` tag. (Basic HTML border attributes are used for visual structural verification).
- Add a descriptive table caption using `<caption>`.
- Define the header block inside `<thead>`. Include column headers for Day/Time and time slots (e.g., 09:00-10:00, 10:00-11:00, 11:00-12:00, 12:00-01:00, 01:00-02:00, 02:00-04:00).
- Add `scope="col"` to all top header cells (`<th>`).

### Step 4: Populating Rows & Merging Cells

- Begin the table body inside `<tbody>` and create rows (`<tr>`) for each day (Monday through Friday).
- Use row headers (`<th scope="row">`) for the day names in the first column of each row.
- Horizontal Merge (`colspan`): For the 12:00 PM - 01:00 PM slot on Monday, create a cell with `colspan="5"` (or appropriate span) to denote "LUNCH BREAK / RECESS" spanning across all days, OR merge across columns for special institutional events.
- Vertical Merge (`rowspan`): Identify a multi-hour lab session (e.g., "Multimedia Systems Lab" on Wednesday from 02:00 PM to 04:00 PM) and use `rowspan="2"` across rows where applicable.

### Step 5: Table Footer Summary

- Create a `<tfoot>` section at the end of the table.
- Add a summary row calculating total lecture hours, lab hours, and total credit points. Merge non-relevant columns using `colspan`.

### Step 6: Verification

- Open timetable.html in a web browser (Chrome, Firefox, or Edge).
- Verify that cell merging does not cause table distortion, stray cells, or broken column alignment.

**Note on Cell Spanning:** When using `colspan="N"`, you are replacing N individual cells in that row. When using `rowspan="N"`, you must omit the corresponding cell in subsequent rows that fall under the spanned area!

## 5. Starter Code Template

Students should start with the following template and edit the sections according to the problem requirements:

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>University Weekly Timetable</title>
</head>
<body>

    <h1>Department of Computer Science & Engineering</h1>
    <h2>Academic Timetable - Semester IV (Session 2026)</h2>
    <p><strong>Cohort:</strong> B.Tech / UG Minor in Web Technologies</p>

    <table border="1" cellpadding="8" cellspacing="0" width="100%">
        <caption><strong>Weekly Course Schedule & Lab Allocation</strong></caption>

        <!-- TABLE HEADER -->
        <thead>
            <tr align="center" bgcolor="#e2e8f0">
                <th scope="col">Day / Time</th>
                <th scope="col">09:00 - 10:00 AM</th>
                <th scope="col">10:00 - 11:00 AM</th>
                <th scope="col">11:00 - 12:00 PM</th>
                <th scope="col">12:00 - 01:00 PM</th>
                <th scope="col">01:00 - 02:00 PM</th>
                <th scope="col">02:00 - 04:00 PM</th>
            </tr>
        </thead>

        <!-- TABLE BODY -->
        <tbody align="center">
            <!-- MONDAY -->
            <tr>
                <th scope="row" bgcolor="#f1f5f9">Monday</th>
                <td>Data Structures</td>
                <td>Digital Logic</td>
                <td>Web Fundamentals</td>
                <!-- TODO: Add Lunch Break Spanning or Individual Slot -->
                <td>Lunch Break</td>
                <td>Mathematics III</td>
                <td>Library / Mentoring</td>
            </tr>

            <!-- TUESDAY -->
            <tr>
                <th scope="row" bgcolor="#f1f5f9">Tuesday</th>
                <td>Digital Logic</td>
                <td>Data Structures</td>
                <td>Mathematics III</td>
                <td>Lunch Break</td>
                <td colspan="2" bgcolor="#fef08a">
                    <strong>Multimedia Systems Lab (Group A)</strong>
                </td>
            </tr>

            <!-- WEDNESDAY -->
            <tr>
                <th scope="row" bgcolor="#f1f5f9">Wednesday</th>
                <td colspan="2" bgcolor="#bbf7d0">
                    <strong>Hardware Workshop (Group B)</strong>
                </td>
            </tr>
        </tbody>
    </table>

```

```

        <td>Web Fundamentals</td>
        <td>Lunch Break</td>
        <td>Data Structures</td>
        <td>Environmental Science</td>
    </tr>

    <!-- THURSDAY & FRIDAY (STUDENTS TO COMPLETE) -->
    <!-- TODO: Implement Thursday and Friday rows using appropriate colspan/rowspan -->
</tbody>

<!-- TABLE FOOTER -->
<tfoot bgcolor="#e2e8f0">
    <tr>
        <th scope="row" colspan="2">Total Summary</th>
        <td colspan="5" align="left">
            <strong>Total Lectures:</strong> 15 Hrs/Week |
            <strong>Total Labs:</strong> 6 Hrs/Week |
            <strong>Total Credits:</strong> 20
        </td>
    </tr>
</tfoot>
</table>

</body>
</html>

```